

Student Career & Employability Analysis 2026

Project Description

Overview

The **Student Career & Employability Analysis 2026** is an open-source, end-to-end Data Analytics project designed to analyze students' career readiness, technical skills, learning habits, and employability.

Unlike traditional analytics projects that rely on publicly available datasets, this project collects **real-world survey data** directly from students using Google Forms.

The project demonstrates the complete data analytics lifecycle, from data collection to business recommendations.

Project Goal

The objective of this project is to identify trends, patterns, and challenges faced by students while preparing for internships and jobs.

The insights generated from this project can help:

- Students
- Colleges
- Placement Cells
- Training Institutes
- Career Mentors

make better data-driven decisions.

Problem Statement

Many students invest significant time in learning technical skills but often lack clarity regarding:

- Which skills are actually in demand?
- How prepared are students for placements?
- What are the biggest employability challenges?
- How confident are students about getting a job?
- Which areas require improvement?

This project aims to answer these questions using survey data and analytics.

Project Objectives

- Design an anonymous survey
- Collect real-world data
- Clean and validate responses
- Perform exploratory data analysis
- Store data in SQL
- Answer business questions using SQL
- Build an interactive Power BI dashboard
- Publish an analytical report

- Provide actionable recommendations

Tech Stack

Data Collection

- Google Forms
- Google Sheets

Data Cleaning

- Python
- Pandas
- NumPy

Database

- PostgreSQL / MySQL

Analysis

- SQL
- Python

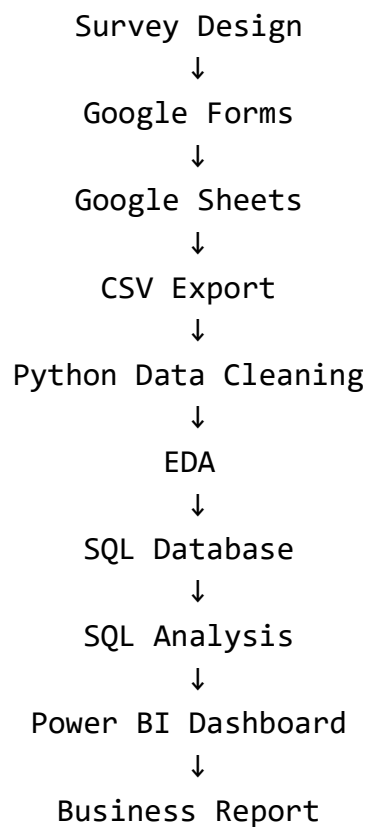
Visualization

- Power BI Desktop

Documentation

- Markdown
- GitHub

Project Workflow



Dashboard Structure

Dashboard 1

Executive Summary

Purpose

Provide a high-level overview of the survey.

KPIs

- Total Responses
- Average Confidence Score
- Resume Adoption %
- LinkedIn Adoption %
- GitHub Adoption %
- Average Study Hours
- Average Projects Completed

Charts

- Career Goal Distribution
- Biggest Challenges
- Confidence Distribution
- Responses by Year of Study

Dashboard 2

Skills Analysis

Purpose

Understand students' technical skills.

Business Questions

- Which technical skills are most common?
- Which skills are students currently learning?
- Which skills have the highest confidence?
- Which skills require more attention?

Charts

- Top Technical Skills
- Skills Currently Learning
- Confidence by Skill
- Skill Distribution by Year

Dashboard 3

Learning Habits

Purpose

Analyze learning behavior.

Business Questions

- How many hours do students study?
- Which learning platforms are most popular?
- Do certifications improve confidence?
- Do students who study more complete more projects?

Charts

- Study Hours
- Learning Platforms
- Certification Status
- Confidence vs Study Hours
- Projects vs Study Hours

Dashboard 4

Career Readiness

Purpose

Evaluate employability.

Business Questions

- How many students have resumes?
- How many students use LinkedIn?
- How many students maintain GitHub portfolios?
- How many projects have students completed?
- How many students have applied for jobs?
- How many interviews have students attended?

Charts

- Resume Status
- LinkedIn Status
- GitHub Adoption
- Projects Completed
- Job Applications
- Interviews Attended

Dashboard 5

Career Aspirations

Purpose

Understand career goals.

Business Questions

- Which career path is most popular?
- What salary do students expect?
- How confident are students?
- Which year has the highest confidence?

Charts

- Target Career
- Expected Salary
- Confidence Score
- Confidence by Year

Dashboard 6

Challenges & Recommendations

Purpose

Identify employability gaps.

Business Questions

- What is the biggest challenge?
- Which areas need improvement?
- Which recommendations should colleges consider?

Charts

- Biggest Challenge
- Improvement Areas
- Open Response Themes
- Word Cloud (Optional)

SQL Analysis Questions

Respondent Overview

- Total number of responses
- Responses by year
- Responses by field
- Responses by state

Skills

- Most common skills
- Skills currently being learned
- Average confidence by skill
- Skills by field of study

Learning

- Average study hours
- Most used learning platform
- Certification distribution
- Study hours vs confidence

Career Readiness

- Resume adoption %
- LinkedIn adoption %
- GitHub adoption %
- Projects completed
- Job applications
- Interview count

Career Goals

- Most preferred career
- Expected salary distribution
- Confidence distribution

Challenges

- Biggest challenge
- Improvement areas

- Frequently mentioned concerns

Python Tasks

Data Cleaning

- Import CSV
- Rename columns
- Remove duplicates
- Handle missing values
- Standardize responses
- Validate data

Exploratory Data Analysis

- Summary statistics
- Frequency analysis
- Correlation analysis
- Distribution analysis
- Missing value analysis

Feature Engineering

- Split multiple skills
- Create confidence categories
- Create employability score
- Create project experience groups

Power BI Features

- Interactive Slicers
- Drill-through Pages
- Tooltips
- KPI Cards
- Bookmarks
- Navigation Buttons
- Dynamic Titles
- DAX Measures

- Conditional Formatting

Expected Insights

Examples include:

- Students with GitHub portfolios tend to report higher confidence.
- Resume adoption is significantly higher than GitHub adoption.
- Most students rely on YouTube for learning.
- A large percentage of students have completed few or no projects.
- Interview experience remains limited despite many job applications.
- Communication and lack of practical projects emerge as common barriers.

Deliverables

- Google Form
- Google Sheets Dataset
- Clean CSV
- Python Notebook
- SQL Database
- SQL Query File
- Power BI Dashboard (.pbix)
- PDF Analytics Report
- GitHub Repository
- README Documentation
- Project Presentation

Target Audience

- Students
- Fresh Graduates
- Colleges
- Placement Cells
- Career Coaches
- Recruiters
- Data Analytics Learners

Skills Demonstrated

Analytics

- Data Collection
- Data Cleaning
- Data Validation
- Data Transformation
- Exploratory Data Analysis

SQL

- SELECT
- WHERE
- GROUP BY
- HAVING
- CASE
- CTE
- Window Functions
- Joins
- Aggregate Functions

Python

- Pandas
- NumPy
- Matplotlib
- Data Cleaning
- Feature Engineering
- Statistical Analysis

Power BI

- Data Modeling
- DAX
- Power Query
- Dashboard Design
- Data Storytelling
- Business Intelligence

Soft Skills

- Survey Design
- Problem Solving
- Business Thinking
- Data Storytelling
- Documentation
- Communication

Open-Source Vision

This project is designed to be more than a portfolio piece. It serves as a complete learning resource for aspiring data analysts. By documenting every stage from survey creation and data collection to SQL analysis, Python workflows, Power BI dashboards, and reporting others can reproduce the project, contribute improvements, or adapt it to different domains such as healthcare, education, HR, or retail.